

1 Context and consent

Thank you for taking the time to answer this questionnaire and take part in this study. The purpose of this study is to understand the structural robustness of global land use and hydrological models, focusing on how software architecture relates to model stability and maintenance risk. The data collected in this questionnaire will be used to understand how the experience of developers and maintainers of **[the model]**'s code compares to our network-based analysis of **[the model]**'s call graph.

All data will be gathered, stored and protected in line with GDPR. Your answers will remain anonymous, we will not collect any names, IP or email addresses, and your participation is voluntary. As your answers to this questionnaire are anonymous, withdrawal from the questionnaire can only be done by exiting the tab being used before the questionnaire is submitted. Once submitted, data cannot be withdrawn as it will not be identifiable.

By selecting "yes" to the below "I consent" question, you:

1. Understand that participation in this questionnaire is completely voluntary.
2. Give permission for your anonymized answers to be used within this study and academic works.
3. Understand and agree to the fact that due to the anonymity of this questionnaire, once the questionnaire is submitted, your data cannot be withdrawn.
4. Understand that your data will be gathered, stored and protected in line with GDPR

I consent to the above: **[yes check box]**

2 Background

2.1 Your role in relation to the models

- Core maintainer.
- Contributor.
- User.
- (Other). Please specify.

2.2 Your familiarity with the model codebase

(1 = not familiar, 5 = very familiar)

3 Path-level evaluation

For each of the 10 paths:

How familiar are you with the following call sequence?

A → B → C → D

- 0 - Not familiar at all.
- 1 - Aware of some functions, but not the path.
- 2 - Moderately familiar.
- 3 - Very familiar.

If the respondent answers 0, we skip evaluations for this path and move to the next.

3.1 Evaluation questions

1. The path corresponds to a real execution chain in a typical model run.
2. The path traverses routines that are central to the model's main workflow.
3. The path is visited often by developers for bug fixes, refactoring or adjustments.
4. If a fault were to occur in the path, it would plausibly have system-wide consequences (e.g., crashes, widespread anomalies).

Possible answers:

- 1 - Strongly disagree
- 2 - Disagree
- 3 - Neither agree nor disagree
- 4 - Agree
- 5 - Strongly Agree

We create a box for comments by the maintainer in each path-level evaluation.